National University of Computer and Emerging Sciences

Lab Manual 01

Programming Fundamentals

Course Instructor	Ms. Anosha Khan
Lab Instructor	Muhammad Hasan
Lab Instructor Email ID	m.hasan@lhr.nu.edu.pk
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FAST - School of Computing
FAST-NU, Lahore Campus

Activities:

Activity 01:

Use of Google Classroom, Flex and Servers (Cactus, Xeon & Exam) is demonstrated by instructor.

Activity 02: Basic Logic Base Problems

1. Using only a 4-minute and 7-minute hourglass timer how would you measure exactly 9 minutes?
2. There are 100 light bulbs lined up in a row in a long room. Each bulb has its own switch and is currently switched off. The room has an entry door and an exit door. There are 100 people lined up outside the entry door. Each bulb is numbered consecutively from 1 to 100. So is each person. Person No. 1 enters the room, switches on every bulb, and exits. Person No. 2 enters and flips the switch of every second bulb (turning off bulbs 2, 4, 6...). Person No. 3 enters and flips the switch of every third bulb (changing the state on bulbs 3, 6, 9...). This continues until all 100 people have passed through the room.
   What is the final state of bulb No. 64? And how many of the light bulbs are illuminated after the 100th person has passed through the room?
   Hint:
   a. Each bulb is toggled (switched on or off) by a person only if the bulb number is a multiple of that person's number.
   b. Number is a perfect square only if it has an odd number of divisors.
3. Three humans, one big monkey and two small monkeys are to cross a river: Only humans and the big monkey can row the boat. At all times, the number of humans on either side of the river must be greater or equal to the number of monkeys on that side. (Or else the humans will be eaten by the monkeys!)
   The boat only has room for 2 (monkeys or humans). Monkeys can jump out of the boat when it's banked.
4. Swap the frogs such that 3 from the left have to jump on the 3 stones on the right and vice versa. Each frog can jump just on the adjacent stone or jump over another frog if there is an empty stone behind it.
   Initial Positions:
   L1, L2, L3, _, R1, R2, R3

Activity 03: Computational Problems

Write an algorithm and draw a flow chart for the following statements.

1. Write a solution that inputs radius from user, calculates the area of circle and displays result on the screen. (Hint: Area of Circle = 3.14 * radius * radius)
2. Cheezious needs a procedure to calculate the number of slices a pizza of any size can be divided into. The cook knows the diameter of the pizza in inches, and each slice should have an exact area of 14.125 inches.
3. A company has a system that pays salary to its workers on weekly basis. The manager at the end of every week must do lengthy manual work to calculate the salary of each worker. The employee is to be paid their basic wage for first 35 hours worked and overtime pay for all hours above 35 on the assumption that the base pay rate is 20.7 while overtime pay rate is 30.5. Now the company wants to automate the process. The input includes the total hours worked by a worker this week and the output includes the total pay for the week.
4. To make a profit, the prices of the items sold in a furniture store are marked up by 80%. After marking up the prices each item is put on sale at a discount of 10%. Design a solution to find the

selling price of an item sold at the furniture store. What information do you need to find the selling price?

5. Suppose that the cost of sending an international fax is calculated as follows:
   The service charge is $3.00, $.20 per page for the first 10 pages, and $0.10 for each additional page. Design a solution that asks the user to enter the number of pages to be faxed and calculates the amount due.
6. An ATM allows a customer to withdraw a maximum of $500 per day. If a customer withdraws more than $300, the service charge is 4% of the amount over $300. If the customer does not have sufficient money in the account, the ATM informs the customer about the insufficient funds and gives the customer the option to withdraw the money for a service charge of $25.00. If there is no money in the account or if the account balance is negative, the ATM does not allow the customer to withdraw any money. If the amount to be withdrawn is greater than $500, the ATM informs the customer about the maximum amount that can be withdrawn. Design a solution that allows the customer to enter the amount to be withdrawn. The algorithm then checks the total amount in the account, dispenses the money to the customer, and debits the account by the amount withdrawn and the service charges, if any.